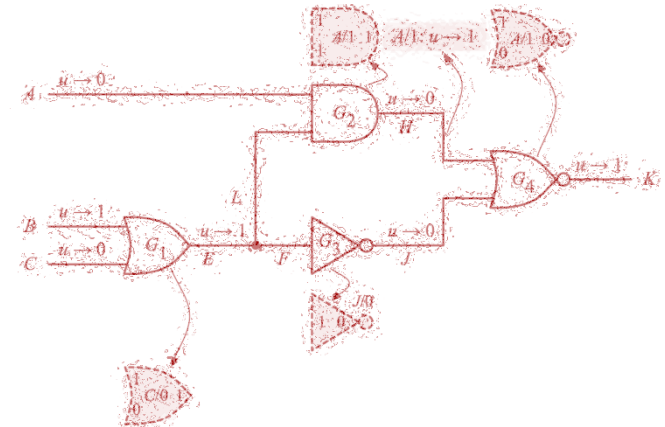


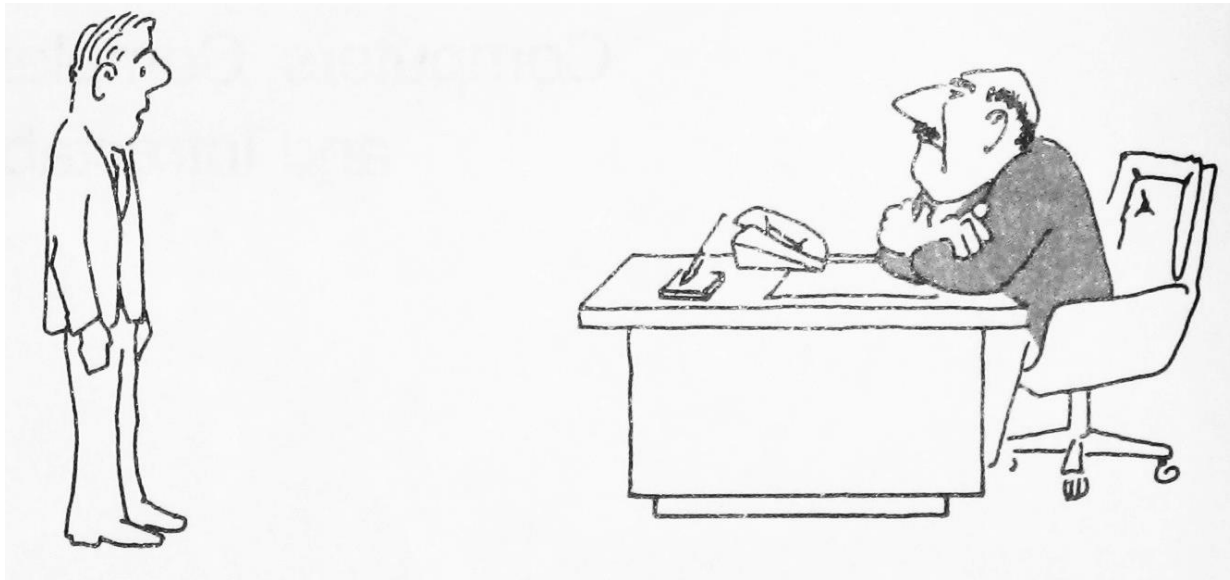
Fault Simulation

- Introduction
- Fault simulation techniques
- Comparison of fault simulation
- Alternatives to fault simulation
 - ◆ Toggle coverage
 - ◆ Fault sampling (1974)
 - ◆ Critical path tracing (1979)
- Issues of fault simulation
- Concluding remarks



Motivating Problem

- Boss: **Why can't we speed up design sign-off?**
- You: But the fault simulation is slow...
- Boss: **We do not need exact fault coverage. Just give me an estimation and DO IT FAST!**



What Should You Do?

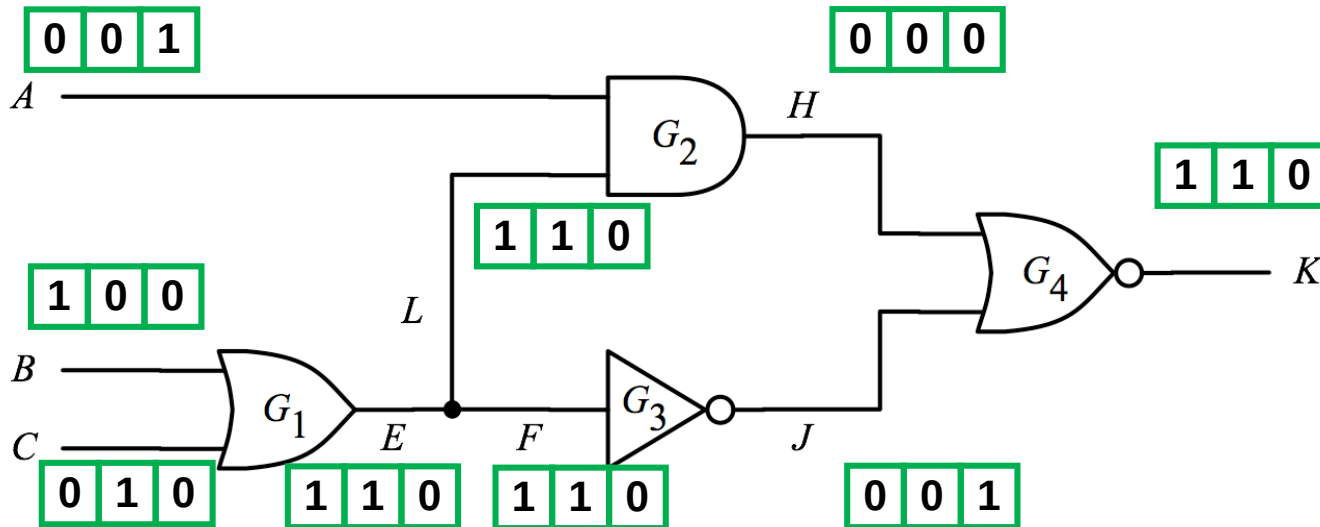
Alternatives to Fault Simulation

- **Q: Is exact fault simulation necessary?**
 - ◆ Exact fault simulation is too expensive
 - ◆ Sometimes, we just want to know *approximate* fault coverage
 - * with reasonable error
- **Approximation is good for**
 - ◆ DFT check in early design phase
 - ◆ Functional test pattern evaluation
- **Approximation is NOT good for**
 - ◆ ATPG
 - ◆ Diagnosis

Toggle Coverage (DEF-1)

$$\text{toggle coverage} = \frac{\sum_{\text{all nodes } i} \# \text{ of different values in node } i}{2 \times \text{total \# of nodes}}$$

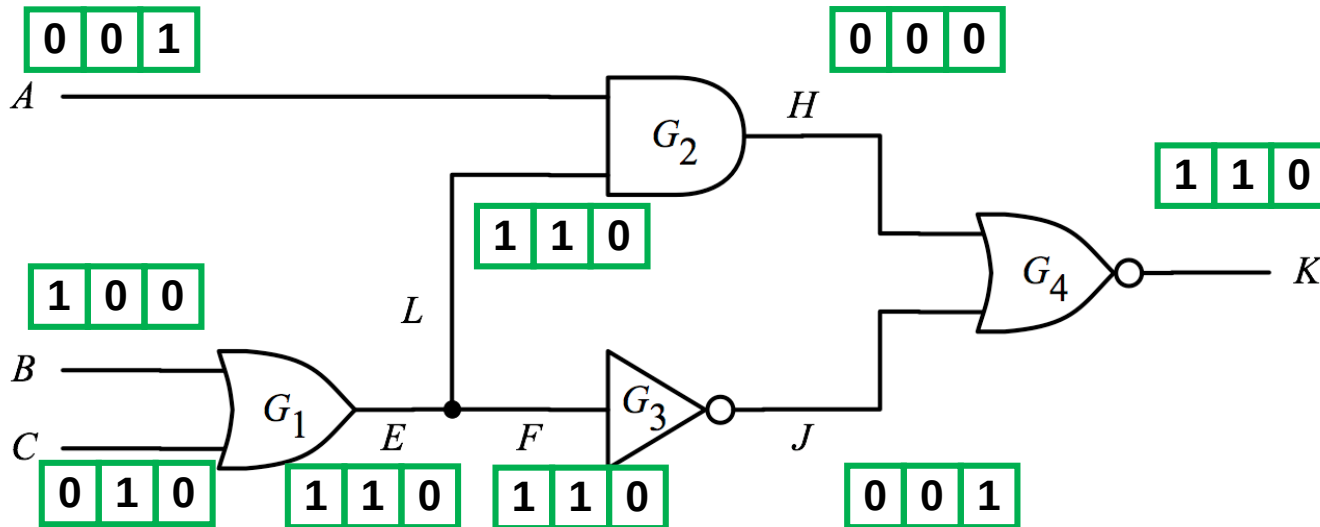
= numbers



Toggle Coverage = 17/18 = 94%

Toggle Coverage

- 😊 Advantage: Toggle coverage is easy to obtain
 - ♦ Logic simulation only, NO fault simulation
- 😞 Disadvantage: Toggle coverage is very optimistic
 - ♦ Fault activation only, NO fault propagation



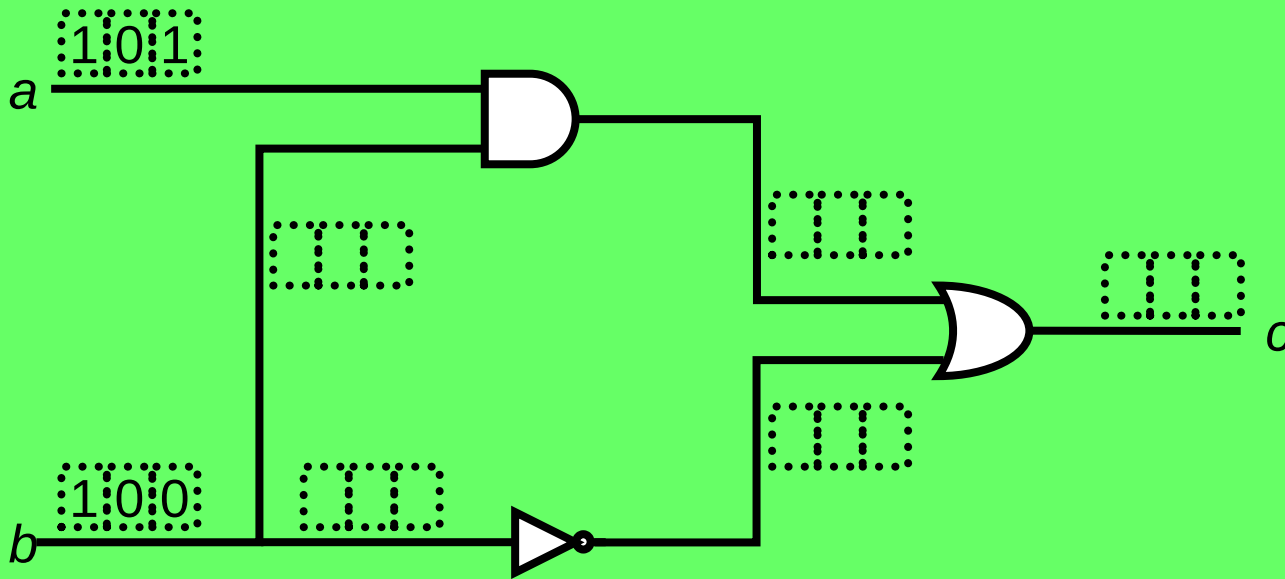
TC is Upper Bound of FC

Fault Coverage = 11/18 = 61% see 5.3 P.10

Toggle Coverage = 17/18 = 94%

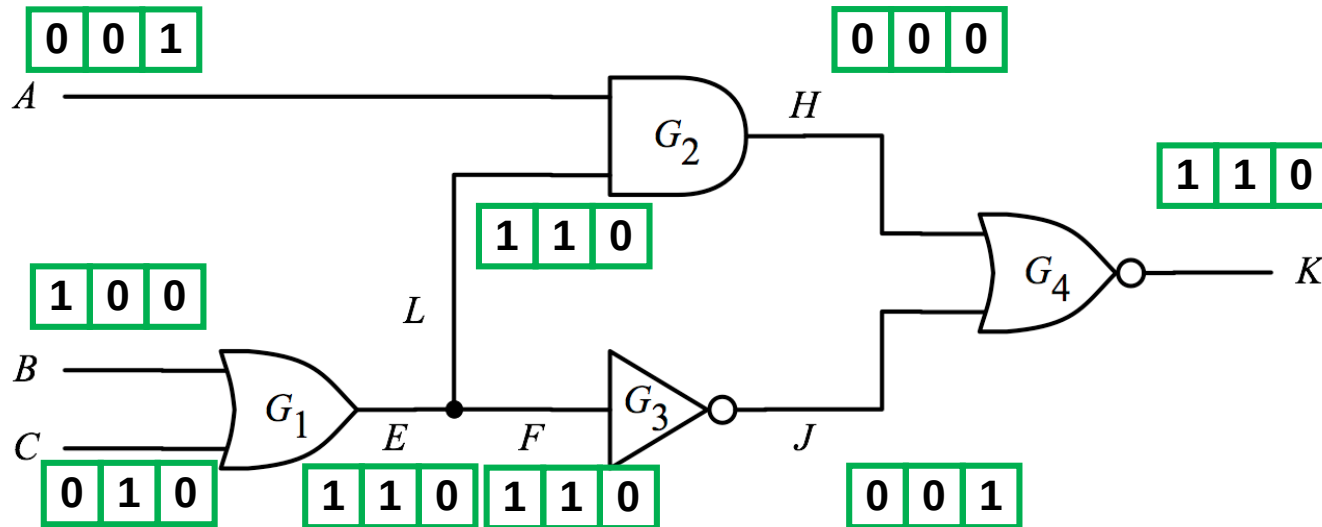
Quiz

Q: Apply 3 patterns to this circuit of 7 nodes. Toggle coverage =?
A:



Toggle Coverage (DEF-2)

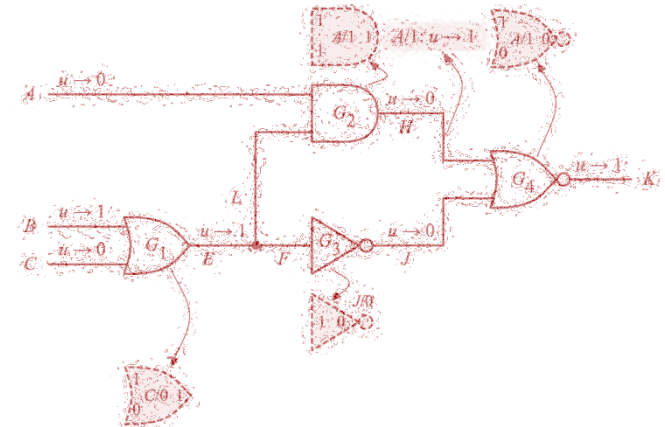
$$\text{toggle coverage} = \frac{\sum_{\text{all nodes } i} \# \text{ of different } \underline{\text{transitions}} \text{ in node } i}{2 \times \text{total \# of nodes}}$$



Toggle Coverage = 9/18 = 50%
DEF-2 more stringent

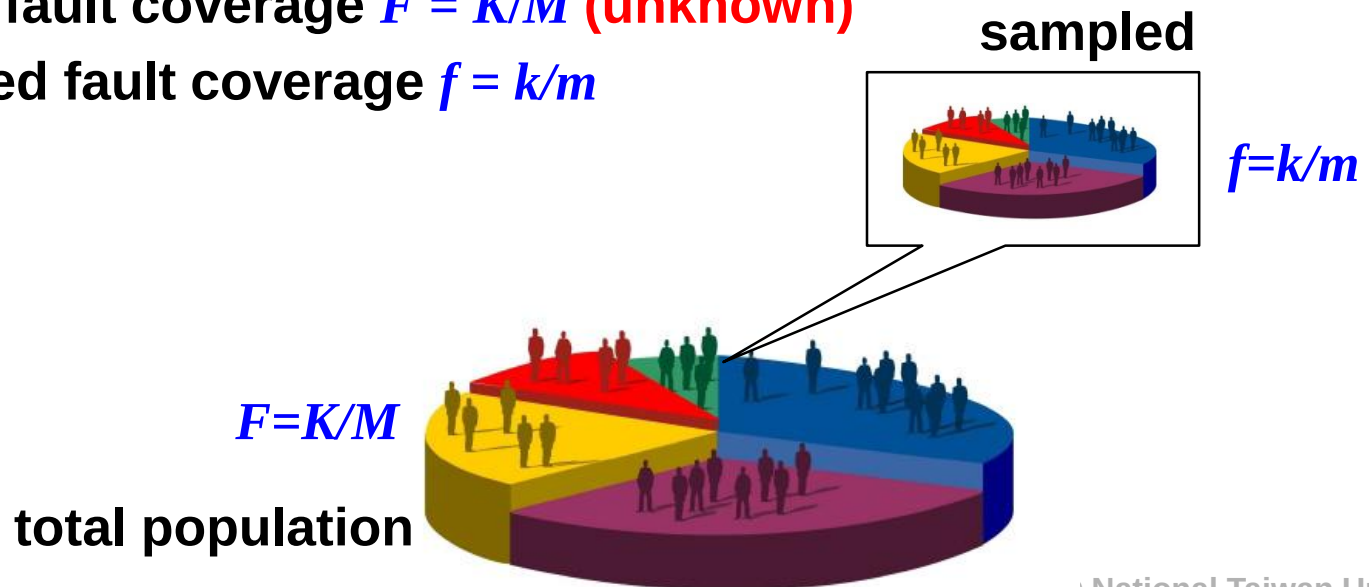
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Fault Sampling [Butler 74]

- Idea: Actual fault simulation of whole circuit is too slow
 - ♦ Can we just sample a small portion of faults?
 - ♦ Like *Polling* before election
- Notation
 - ♦ M : number of total faults
 - ♦ m : number of sampled faults
 - ♦ K : number of total faults detected (**unknown**)
 - ♦ k : number of sampled faults detected
 - ♦ Actual fault coverage $F = K/M$ (**unknown**)
 - ♦ Sampled fault coverage $f = k/m$



Fault Sampling (2)

- $P_k(M, m, K)$ = probability that a test detects k faults from a random sample size of m , given that it detects K faults from M total faults

- ♦ P_k is **hypergeometric distribution** (discrete valued)

$$P_k(m, M, K) = \frac{C_k^K C_{m-k}^{M-K}}{C_m^M}$$

- For large M ,
 - ♦ P_k can be approximated by **normal distribution** (continuous valued)
 - ♦ with mean μ_k and standard deviation σ_k

$$\mu_k = m \frac{K}{M} = mF$$

$$\sigma_k^2 = m \frac{K}{M} \left(1 - \frac{K}{M}\right) \frac{M - m}{M - 1} \cong mF(1 - F) \left(1 - \frac{m}{M}\right)$$

See papers for detailed derivation

Fault Sampling (3)

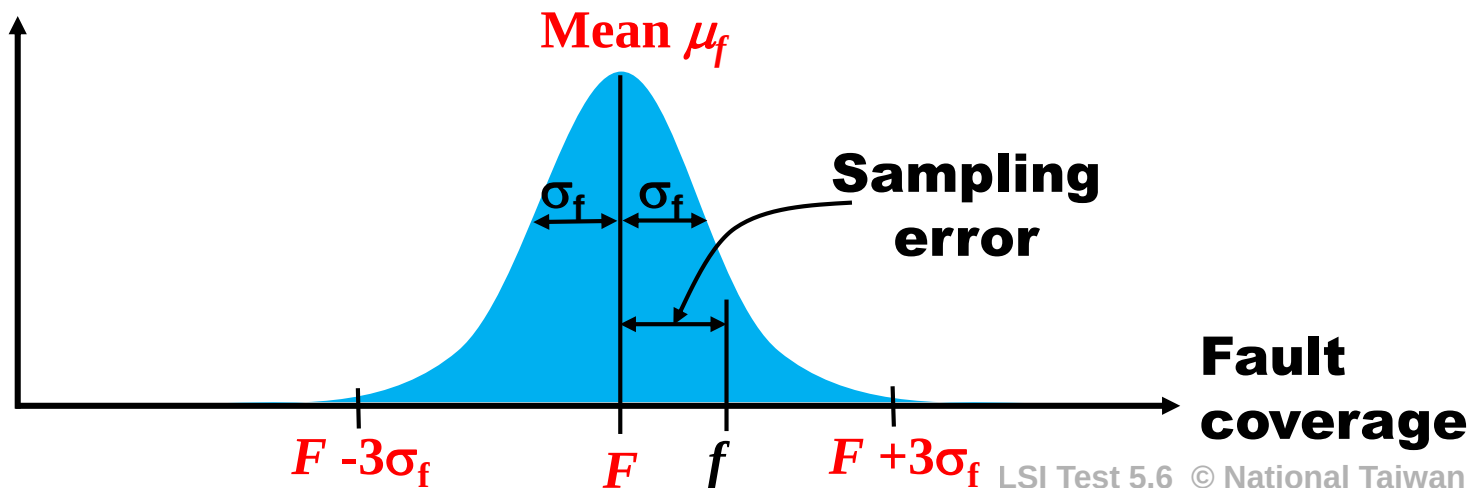
- For large M , sampled fault coverage f is random variable with normal distribution:

$$\mu_f = \mu_k / m = F$$

$$\sigma_f^2 = \sigma_k^2 / m^2 = F(1-F)(1-m/M) / m$$

$$\sigma_f = \sqrt{F(1-F)(1-m/M) / m}$$
$$\cong \sqrt{F(1-F) / m} \quad \text{if } M \gg m$$

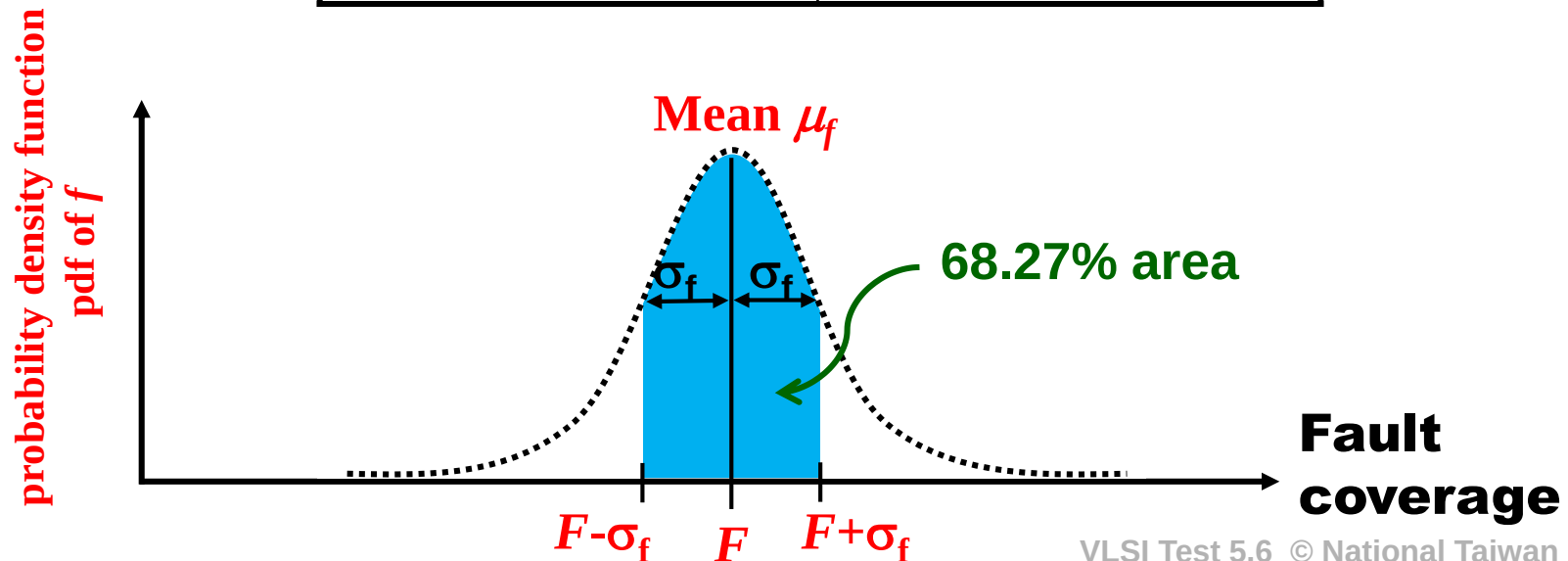
probability density function
pdf of f



Confidence Level

- **Confidence Level** = probability that estimated fault coverage f falls in between the **confidence interval**
- If we assume normal distribution:

Confidence interval	Confidence level
$\mu_f \pm \sigma_f$	68.27%
$\mu_f \pm 2\sigma_f$	95.44%
$\mu_f \pm 3\sigma_f$	99.73%
$\mu_f \pm 4\sigma_f$	99.99%



Example

- $M=1,000,000$
- $m=10,000$ (1% of M)
- NOTE: F is unknown so we use μ_f

$$\sigma_f = \sqrt{F(1-F)(1-m/M)} / m$$
$$\cong \sqrt{F(1-F)} / m$$

$\mu_f (=F)$	70%	80%	90%	95%
σ_f	0.5%	0.4%	0.3%	0.2%
68% CL	70%±0.5%	80%±0.4%	90%±0.3%	95%±0.2%
95% CL	70%±1%	80%±0.8%	90%±0.6%	95%±0.4%
99% CL	70%±1.5%	80%±1.2%	90%±0.9%	95%±0.6%

**Accurate Prediction:
High CL and Small CI**

Quiz

**Q: Suppose FC=95% but we want to improve std σ_f to 0.1%.
How many samples do we need, m =?**

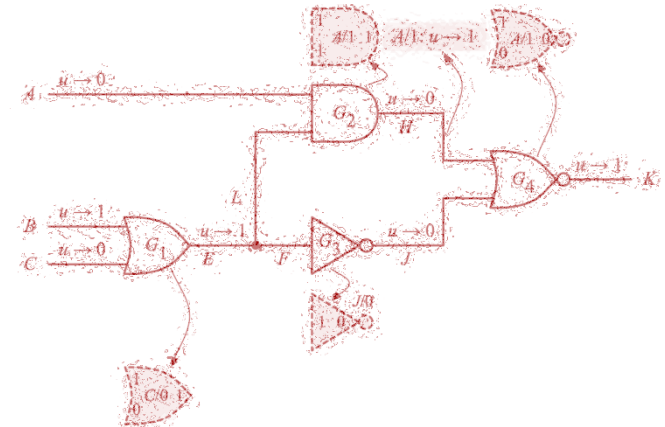
A:

$$\sigma_f = \sqrt{F(1-F)(1-m/M)} / m$$
$$\cong \sqrt{F(1-F)} / m$$

$\mu_f (=F)$	70%	80%	90%	95%
σ_f	0.5%	0.4%	0.3%	0.2%
68% CI	70%±0.5%	80%±0.4%	90%±0.3%	95%±0.2%
95% CI	70%±1%	80%±0.8%	90%±0.6%	95%±0.4%
99% CI	70%±1.5%	80%±1.2%	90%±0.9%	95%±0.6%

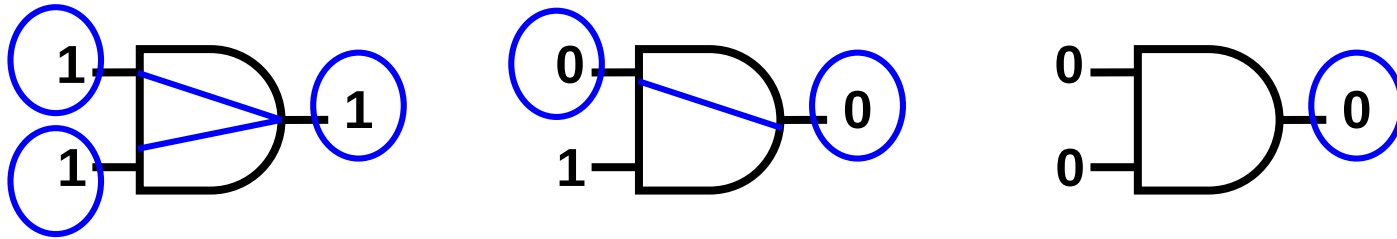
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Critical Path Tracing (CPT) [Roth 79]

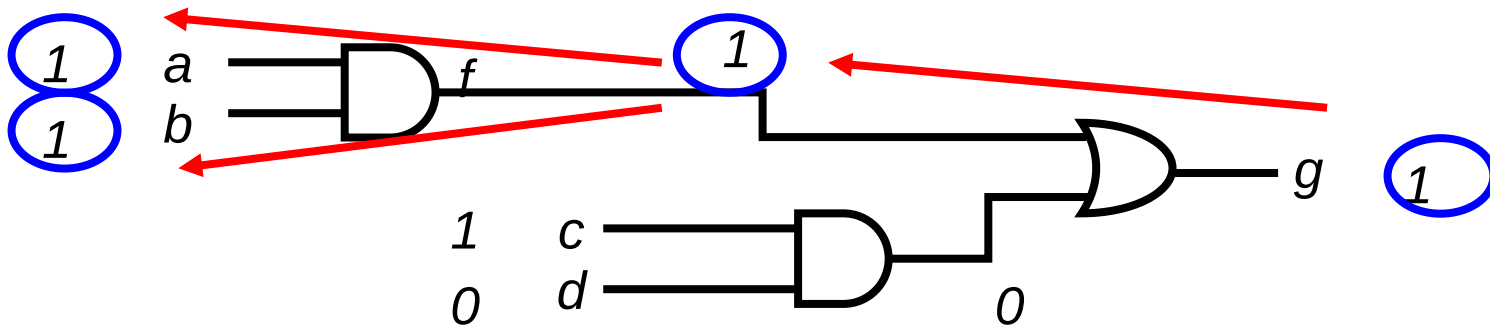
- x is **Critical Signal**
 - ♦ x 's value change causes some **primary output** values to change
- **Critical path**
 - ♦ All signals on this path that are critical signals
- Example: critical signals are circled



Note: In timing analysis, *critical path* has different definition
Critical path = Path with the longest path delay

CPT Algorithm [Abramovici 84]

- **Critical Path Tracing**: Start from primary outputs to primary inputs
 - ♦ If gate **output** is critical, backtrace its critical gate input(s)
- Example: fanout-free cone (FFC) 
 - ♦ Critical signals: *a, b, f, g*. Critical paths: *afg* and *bfg*



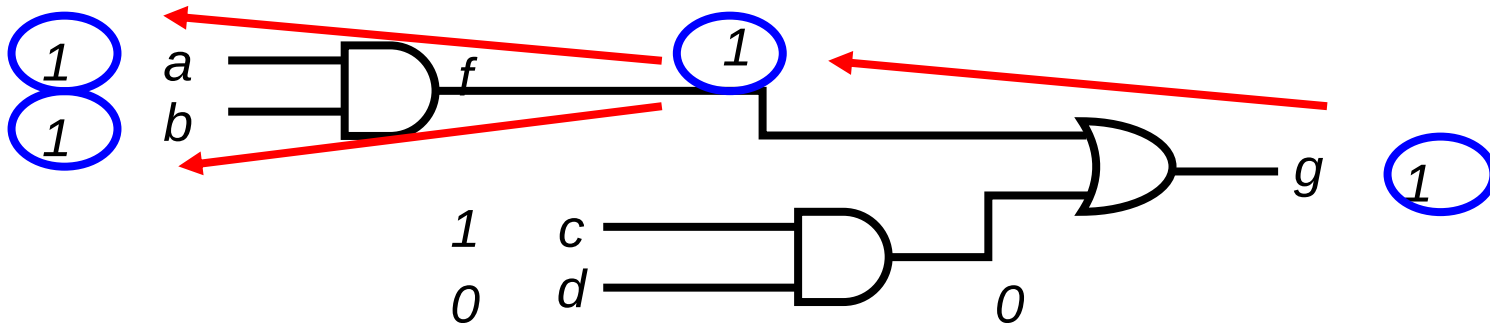
```
CPT (O) /*O is a node*/  
1. foreach gate input g of O  
2.   if (g is critical signal) then  
3.     CPT (g) ;  
4.   else return;
```

CPT Theorem

THEOREM:

If critical signal g has good value v , then g stuck-at v' fault is detected

- Example: Detected faults: a SA0, b SA0, f SA0, g SA0

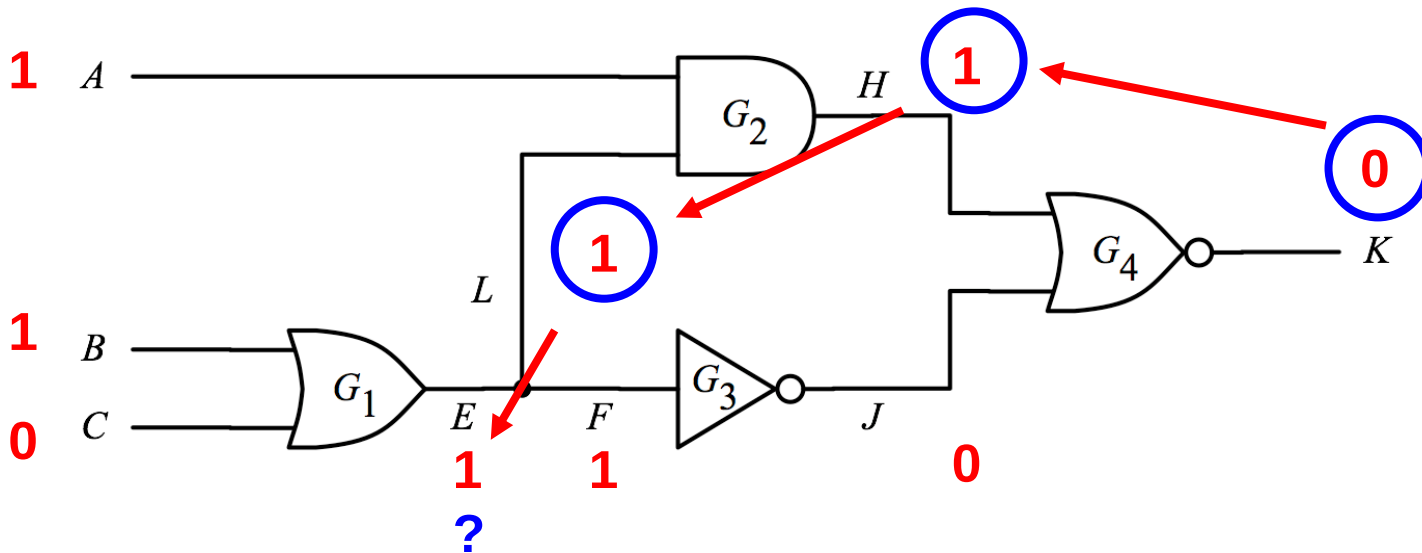


- CPT Advantages:
 - ♦ CPT can be done in **linear time**
 - ♦ Only **logic simulation** needed for FFC
 - ♦ Used in first commercial ATPG tool, LASER

CPT Is Very Fast ... but

How about Fanout Stem?

- L is critical; F is non-critical
- E is **non-critical**

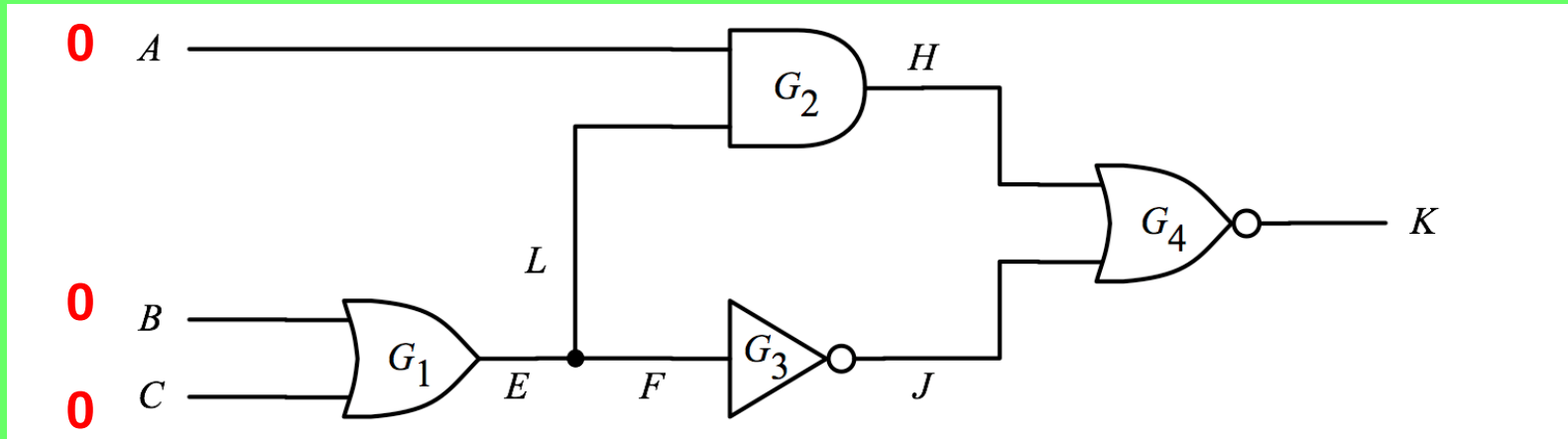


Is This Always The Case?
Fanout stems are non-critical?

Quiz

Q: Determine if fanout stem E is critical or not

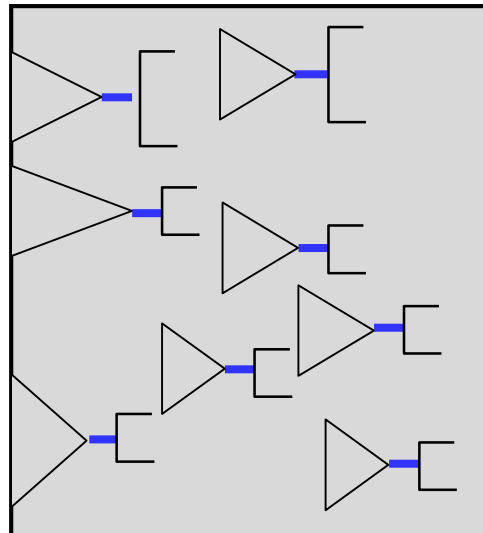
A:



**CPT Does NOT Work Well
when Fanout Reconverges**

So... What Should We Do?

- 1. Count fanout stems as non-critical
 - ♦ Fast but fault coverage pessimistic
- 2. Only fault simulate fanout stems, CPT in fanout-free cones
 - ♦ Slow but fault coverage accurate

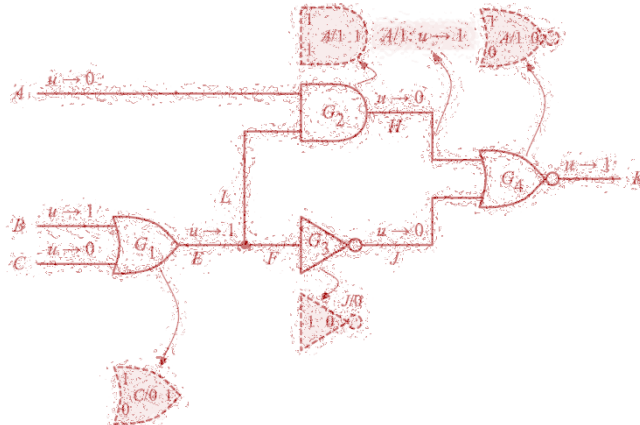


 fanout-free cone
 fanout stem

Summary

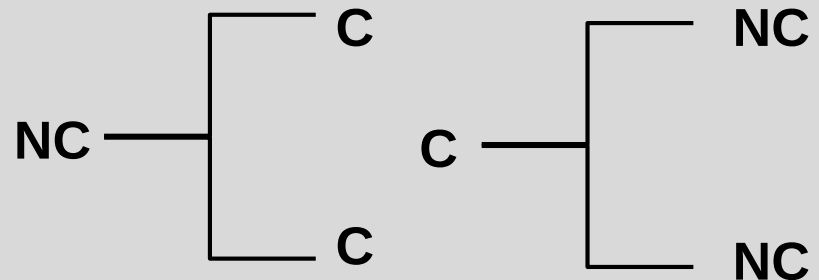
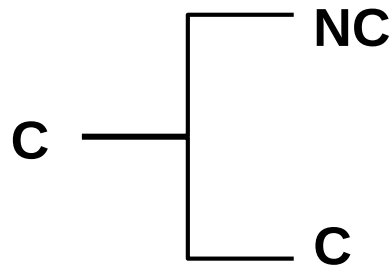
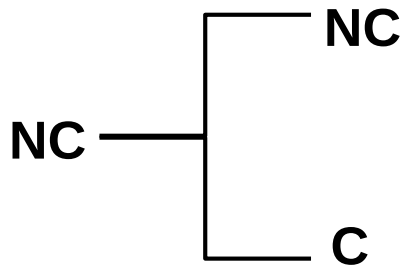
- Alternatives to fault simulation, fast but inaccurate

- ① Toggle coverage
 - ◆ Only **logic simulation** needed
- ② Fault Sampling
 - ◆ Only fault simulate a small portion of faults
 - ◆ **Confidence interval** and **confidence level**
- ③ Critical path tracing
 - ◆ **Linear time** for fanout-free cones
 - ◆ Still need fault simulation when **fanouts reconverge**



FFT

- Q1: For CPT, all four cases are possible.
 - ♦ We have shown first two cases
 - ♦ Find example for latter two cases?



- Q2: Toggle coverage $\geq$ fault coverage. Can you prove it?
 - ♦ assume DEF-1 toggle coverage